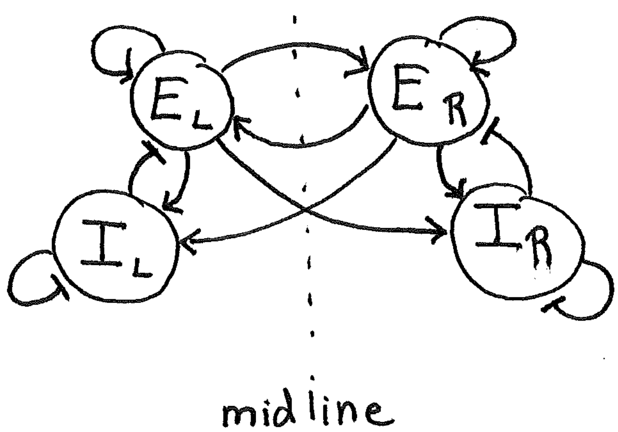
Oculomotor integrator:

When you move your eyes, active motor drive is needed to keep them still. This is accomplished by a distributed neural network in the brainstem and cerebellum that integrates eye velocity commands into the eye position.

$$\underbrace{x(t)}_{\text{position}} = x(0) + \int_0^t dt' \underbrace{v(t')}_{\text{velocity}}$$

The network architecture that achieves this has been extensively studied by Goldman, Seung, Tank and others. Its basic structure is



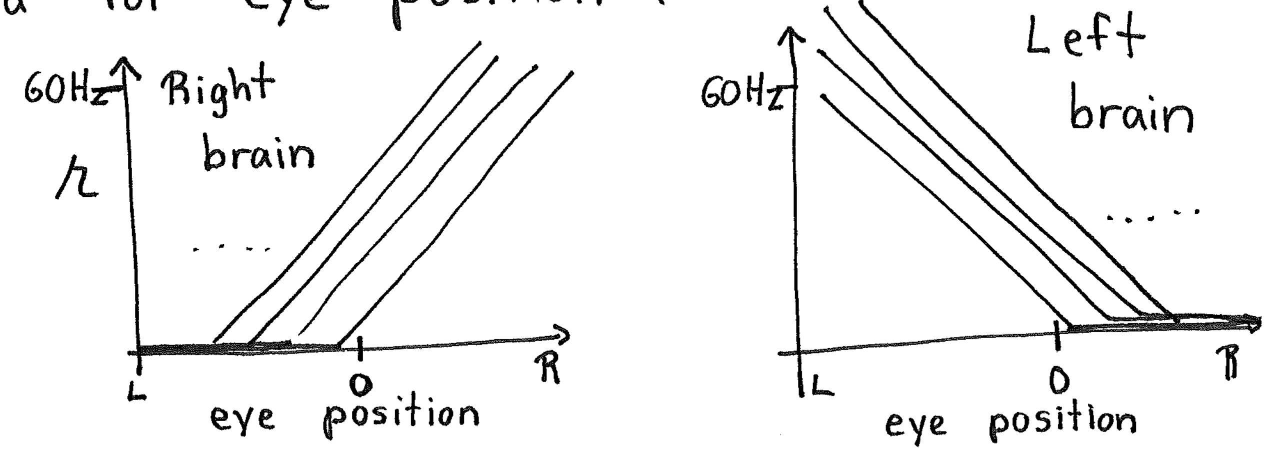
E : excitatory

I : inhibitory

L : Left hemisphere

R : Right hemisphere

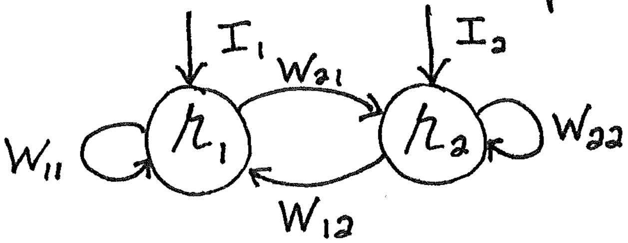
For the network to store the integrated velocity commands with minimal drift (e.g. large τ_{eff}), positive feedback is needed. Neurons are linearly tuned for eye position (Fish data)



Integrator requirements:

Linear tuning curves make the oculomotor integrator amenable to linear neural network analyses.

Two neuron example:



$$W = \begin{bmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{bmatrix}$$

The eigenvalues of W are determined by

$$0 = \det \begin{bmatrix} W_{11}-\lambda & W_{12} \\ W_{21} & W_{22}-\lambda \end{bmatrix} = (W_{11}-\lambda)(W_{22}-\lambda) - W_{12}W_{21}$$

$$= \lambda^2 - \lambda(W_{11}+W_{22}) + W_{11}W_{22} - W_{12}W_{21}$$

$$\lambda_\pm = \frac{1}{2}\left(W_{11}+W_{22} \pm \sqrt{(W_{11}+W_{22})^2 - 4(W_{11}W_{22} - W_{12}W_{21})} \right)$$

A perfect integrator requires $\lambda_+ = 1$, $\lambda_- \leq \lambda_+ = 1$

$$2 - W_{11} - W_{22} = \sqrt{(W_{11}+W_{22})^2 - 4W_{11}W_{22} + 4W_{12}W_{21}}$$

Square both sides

$$4 - 4(W_{11}+W_{22}) + (W_{11}+W_{22})^2 = (W_{11}+W_{22})^2 - 4W_{11}W_{22} + 4W_{12}W_{21}$$

$$W_{12}W_{21} = 1 - W_{11} - W_{22} + W_{11}W_{22} = (1-W_{11})(1-W_{22})$$

If $W_{12}=0$ or $W_{21}=0$, then

$$0 = (1-W_{11})(1-W_{22}) \Rightarrow W_{11}=1 \text{ or } W_{22}=1$$

Self-synapses are required for positive feedback. If not,

$$W_{21} = \frac{(1-W_{11})(1-W_{22})}{W_{12}}$$

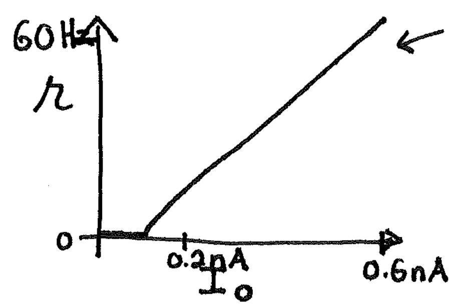
Specifies one parameter in terms of the other three. There is a solution space of synaptic weight matrices that lead to integration.

Parameter variability in biological models:

In 2001, Goldman, Marder, and Abbott showed with simulations that many different parameter settings lead to the same voltage dynamics in a biophysically detailed neuron Model. In the years since, Marder has shown that central pattern generator models also permit parameter variability, and this model variability manifests as biological variability across individual instantiations of the circuit in different animals.

That the oculomotor integrator is approximately linear allowed Goldman to systematically characterize possible weight matrices in 2013.

Measured F-I curve:



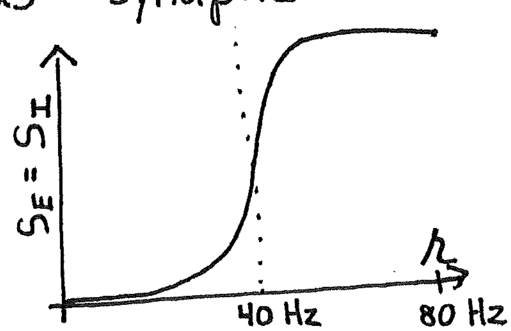
Close to threshold-linear with bias

; $r = f(I_0) \Rightarrow I_0 = f^{-1}(r)$ if $r > 0$

Model of input currents to neurons:

$$\underbrace{f^{-1}(r_i)}_{\substack{\text{current inferred} \\ \text{from firing rate}}} = \sum_{j=1}^{N_E} W_{ij}^{(E)} \underbrace{S_E(r_j^{(E)})}_{\substack{\text{Synaptic nonlinearity} \\ \text{of excitatory inputs}}} + \sum_{j=1}^{N_I} W_{ij}^{(I)} \underbrace{S_I(r_j^{(I)})}_{\substack{\text{Inhibitory} \\ \text{nonlinearity}}} + \underbrace{I_i}_{\substack{\text{Background} \\ \text{input}}}$$

Various synaptic nonlinearities could fit the data. E.g.



(High synaptic threshold)

Fixing the nonlinearity, model fitting reduces to linear regression

$$Y_{i\mu} = g^{-1}(r_i(p_\mu)) \quad ; \quad p_\mu : \mu^{th} \text{ position of eye}$$

$$X_{j\mu} = \begin{cases} s_E(r_j^{(E)}(p_\mu)) , & j = 1, \cdots, N_E \\ s_I(r_{j-N_E}^{(I)}(p_\mu)) , & j = N_E+1, \cdots, N_E+N_I \\ 1 , & j = N_E + N_I + 1 \end{cases}$$

$$\hat{Y}_{i\mu} = \sum_{j=1}^{N_E+N_I+1} \underbrace{W_{ij}}_{\text{Weight AND bias parameter}} X_{j\mu}$$

$$\mathcal{L}(W) = \sum_i \sum_\mu \left(Y_{i\mu} - \hat{Y}_{i\mu}(W) \right)^2$$
$$= \sum_i \mathcal{L}_i(W)$$

Each output neuron i can be fit independently to determine its incoming weights and bias. Letting y and W denote the i^{th} row of the corresponding matrices and $e = \mathcal{L}_i$,

$$\hat{y} = Wx$$

$$e = \tfrac{1}{2}(y - Wx)(y - Wx)^T$$
$$= (yy^T - \underbrace{yx^T}_{u} W^T - W \underbrace{xy^T}_{u^T} + W \underbrace{xx^T}_{H, \text{ Hessian, Second derivative matrix}} W^T)/2$$

$$= ((W - uH^{-1}) H (W - uH^{-1})^T - uH^{-1}u^T + yy^T)/2$$

$$= e_{min} + \tfrac{1}{2}(W - \hat{W}) H (W - \hat{W})^T$$

where

$$e_{min} = \tfrac{1}{2}(yy^T - uH^{-1}u^T)$$

$$\hat{W} = uH^{-1} = \underbrace{yx^T(xx^T)^{-1}}_{1 \times N \text{ (row vector form of result)}}$$

This equation can be better understood in the basis of eigenvectors of the Hessian (symmetric)

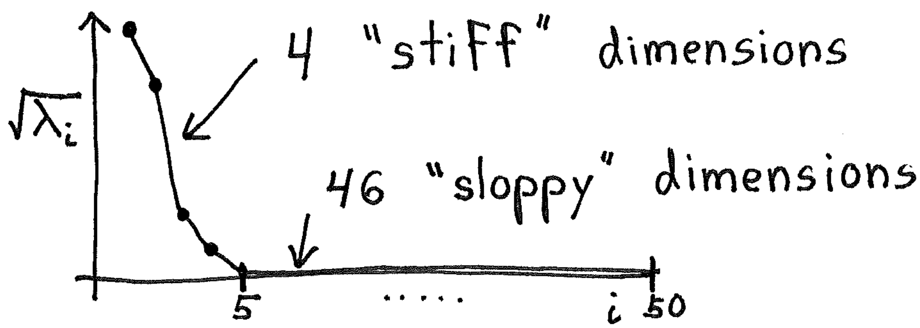
$$H = V D V^T$$

$$e = e_{min} + \frac{1}{2}\left((w-\hat{w}) V D V^T (w-\hat{w})^T\right)$$

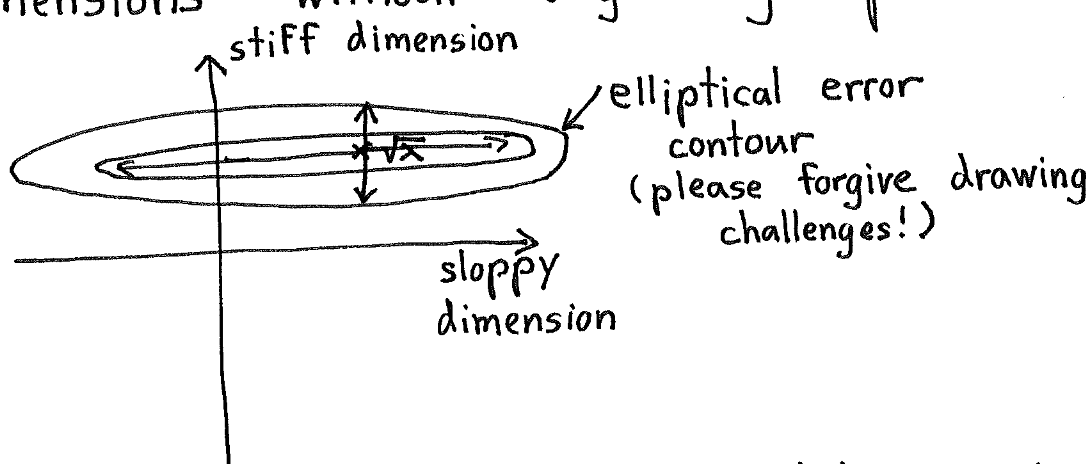
$$= e_{min} + \frac{1}{2}\sum_{j=1}^{N_E+N_I+1} \underbrace{\lambda_j}_{\text{eigenvalue of } H} \underbrace{\left((w-\hat{w})V\right)_j^2}_{\text{deviations from } \hat{w} \text{ along eigenvector direction}}$$

The eigenvalues of H specify how sensitive the model fit is to parameter variations in the direction of its eigenvector.

The oculomotor integrator fit was only sensitive to four dimensions of variation in synaptic weights.



Consequently, the model could vary along a large number of dimensions without degrading performance



It's best to focus on ensembles of models that can adequately capture the data.